Nowcasting Visitor Arrivals (VA) and GDP: Fiji's Experience

IMF STI PEER 2 PEER LEARNING

THE RESERVE BANK OF FIJI



Presentation Outline

ARIMA-X presentation



Background & Context



Technical Aspects



Model Performance



Model Improvements



Communication



Way Forward

Highly Confidential

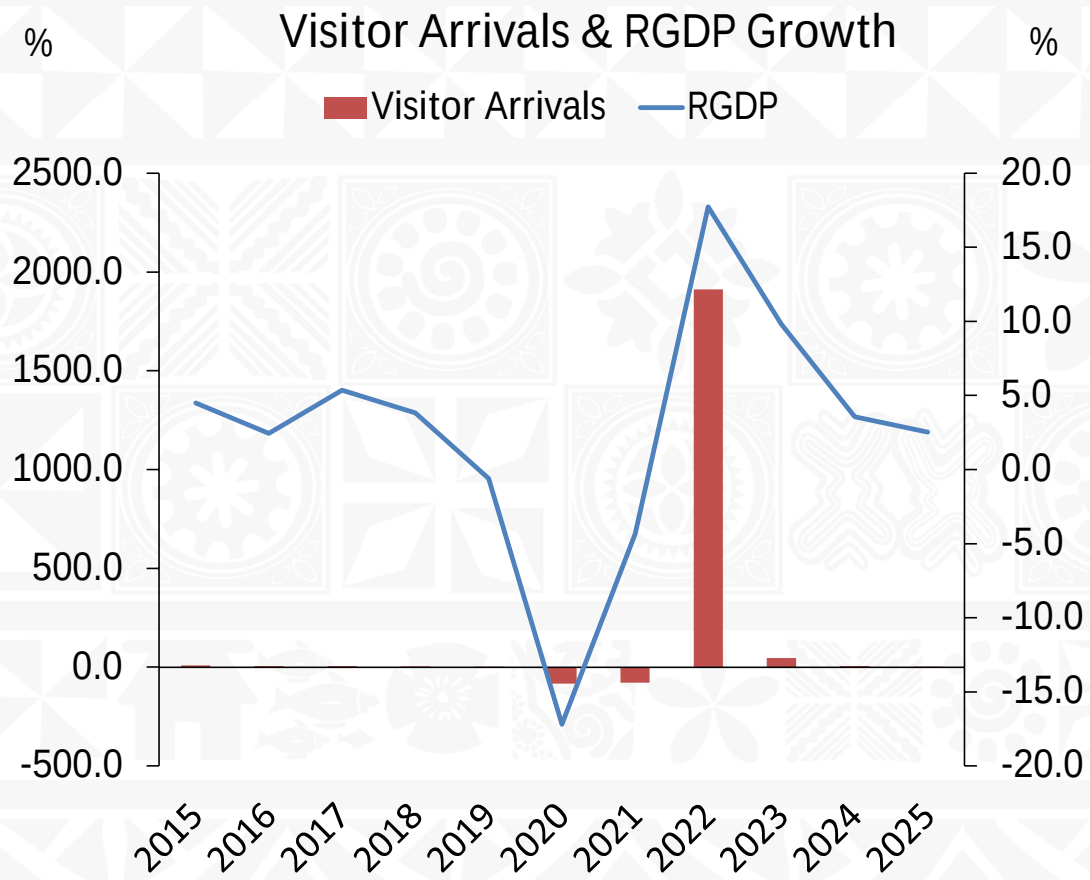
Background & Context

On Fiji's economy

- Tourism contributes roughly around 40% to Fiji's economy

On the VA nowcasting model

- It was developed sometime in 2022 and early 2023
- First used in May 2023
- Used to nowcast monthly visitor arrivals



Highly Confidential

Technical Aspects of the VA Nowcasting Model

On data used:

- Actual visitor arrivals monthly data sourced from the Fiji Bureau of Statistics
- Google Trends data: Search by Fiji flights and hotels searches from key source markets mainly Australia, New Zealand & the United States

On model type:

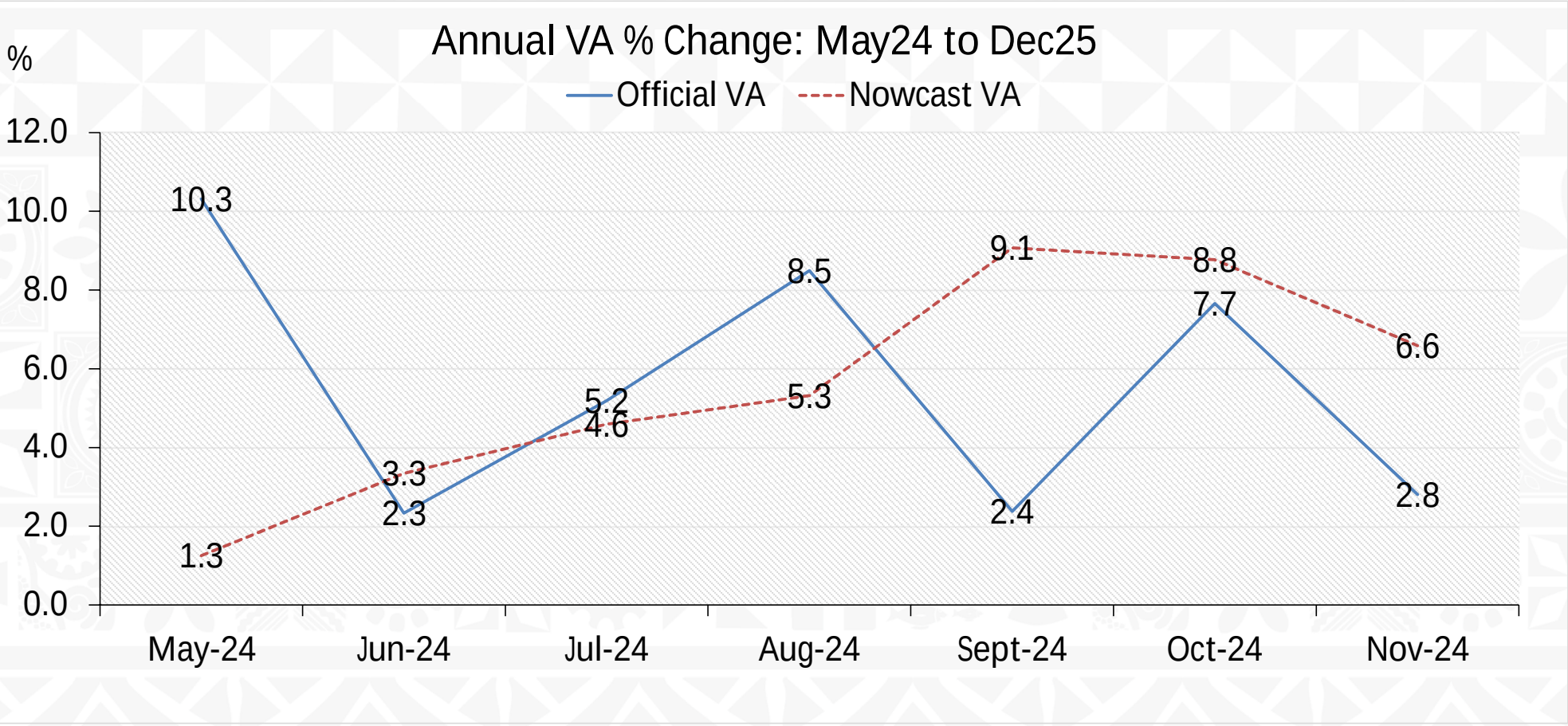
- Employs an ARIMA-X approach, applied through EViews software.
- Predict future values of a time series based on its own past values (lags)

On steps and procedures:

- The exercise is conducted monthly
- Within a month, the nowcasting exercise can be conducted 2-3 times

Highly Confidential

VA Nowcasting Model Performance



Highly Confidential

VA Nowcasting Model Improvements

On data used:

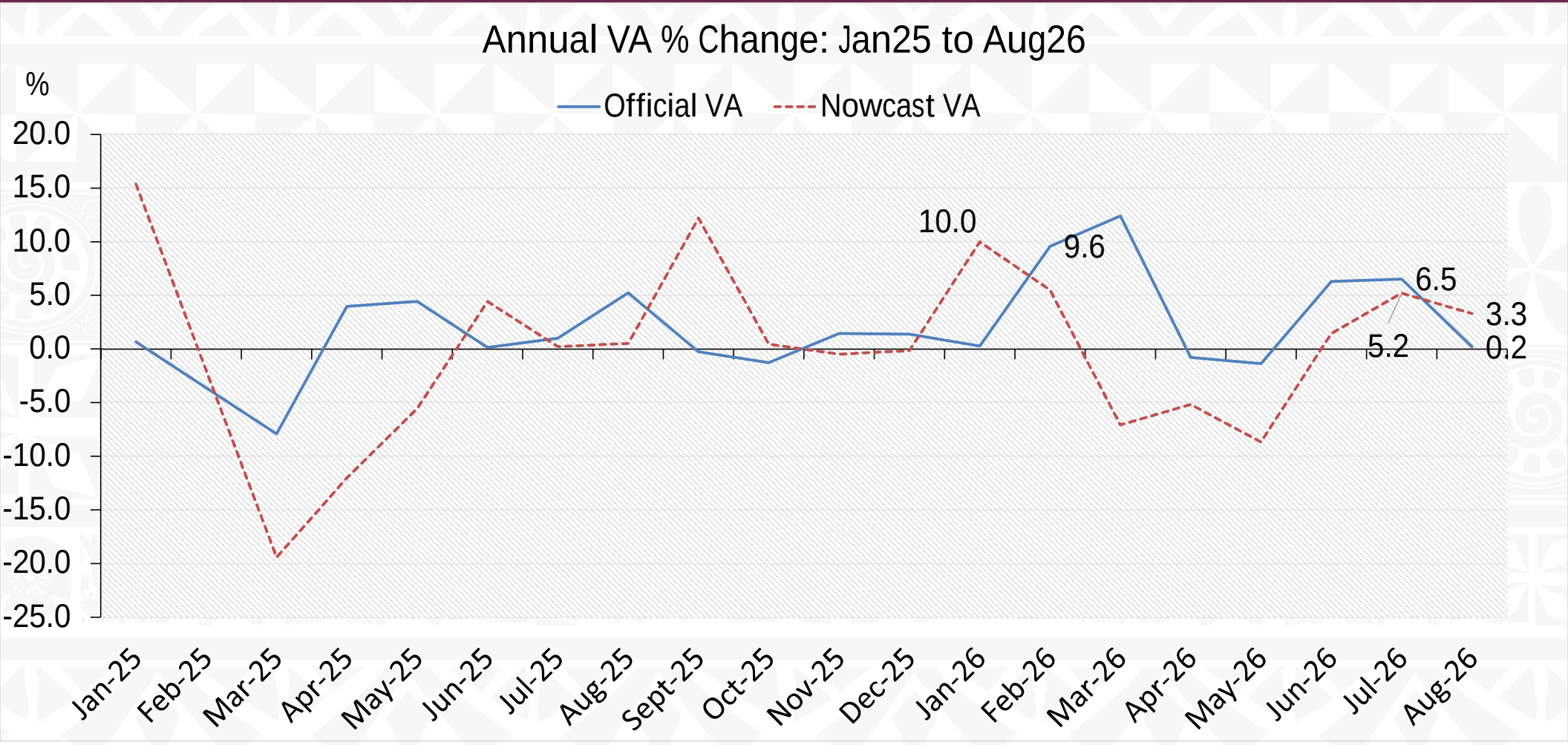
- Initially started off with only using flight searches from Australia
- Have now expanded to include flight from other key source markets mainly Australia, NZ & the US from 2025

Scope of model input:

- Initially started off with only using flight searches
- Have now expanded to include hotel searches i.e. flight & hotel searches from 2025

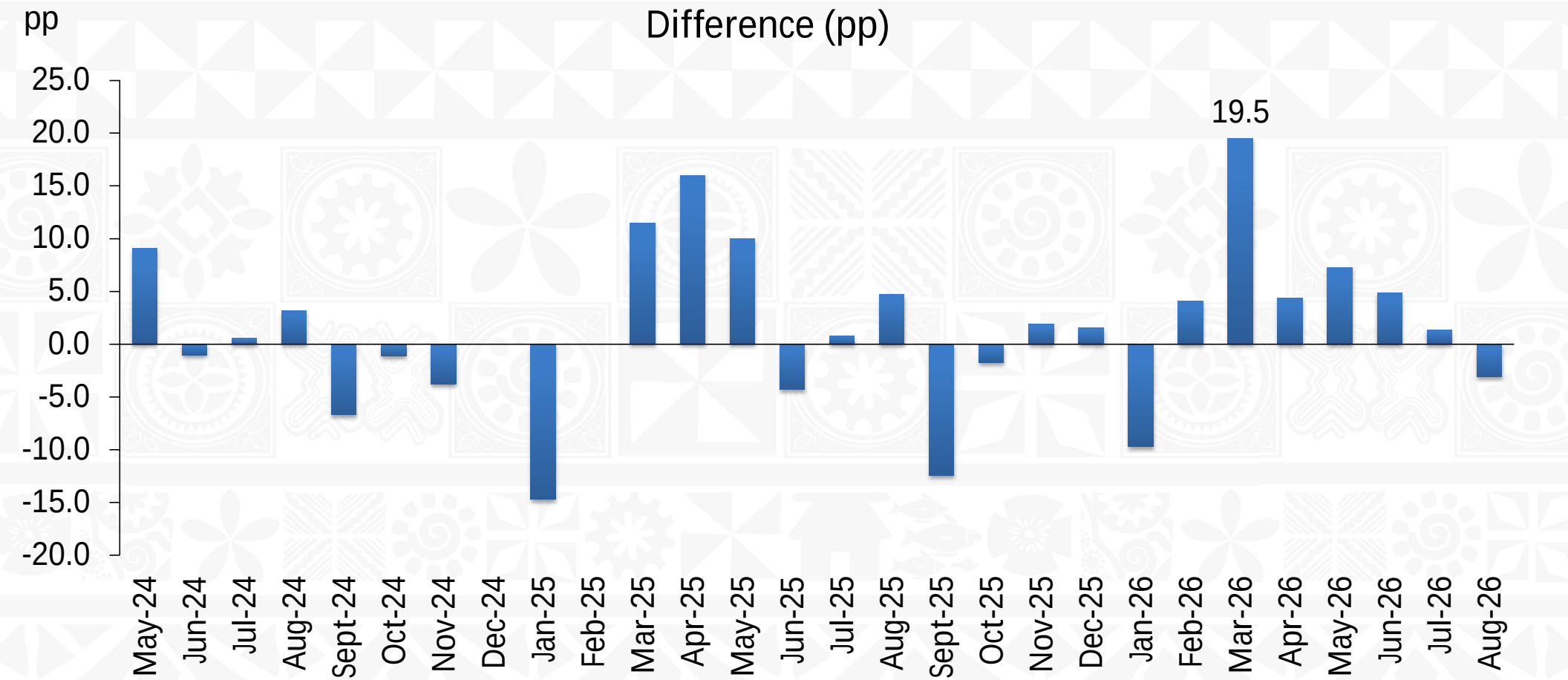
Highly Confidential

VA Nowcasting Model Performance



Highly Confidential

VA Nowcasting Model Performance



Highly Confidential

Communication of the VA Nowcasting Model

Use of VA nowcast model results:

- Presented during policy, monetary policy committee and Board meetings
- A key segment of the information papers to the executive and Board
- Results are deliberated by members part of the associated forums
- Form the basis for macroeconomic forecasting

Industry Outlook

As per the analysis from the Nowcast model,[66] the month of January 2026, is expected to record around 77,873 visitors (10% higher or 7,079 more visitors compared to January 2025). Several events[2] are scheduled in Fiji during the early months of the year, including the Tourism Convention (22-23 January), UB40 Live (30 January), a series of Drua rugby matches (from February to May), Wai Tui International (3-6 April), the South Pacific Tourism Exchange (25-26 March), and the Fiji Fringe Festival (10-18 April). These diverse activities are anticipated to attract travel interest, driving higher visitor inflows, and contributing to the continued expansion of Fiji's tourism industry.

Highly Confidential

Way Forward

Improving data sources & methodology:

- Consider non-traditional data from other sources & search engines
- Seasonally adjusting the historical series to capture cyclical patterns in visitor arrivals
- Accounting for structural break e.g. COVID and Oil crisis
- Consider using the lag of Google trends flights/hotels searches

Stakeholder liaison:

Connecting better with tourism stakeholders to provide more reliable and concrete data beforehand e.g., actual booked flight seats/hotel rooms

Documentation and analysis of nowcast results:

Assess the forecast errors to ensure continuous improvement on the model

Consider assessing the drivers of VA nowcast to facilitate interpreting results

Highly Confidential

The alternative nowcast

Nowcasting Visitor Arrivals Using Tourism Receipts

Highly Confidential

Nowcasting

The model

- Excel based model
- Trend
- Historical coefficients/ratios
  - tourism receipts
  - visitor arrivals
- Allows for value judgement
- Developed to give a range forecast

What this Nowcasting Does

- Uses high-frequency data to estimate monthly visitor arrivals.
- Provides an early indication before official visitor arrival data is released.
- Uses tourism receipts as a proxy indicator for visitor movements.
- *Complements the existing ARIMAX forecasting model.*

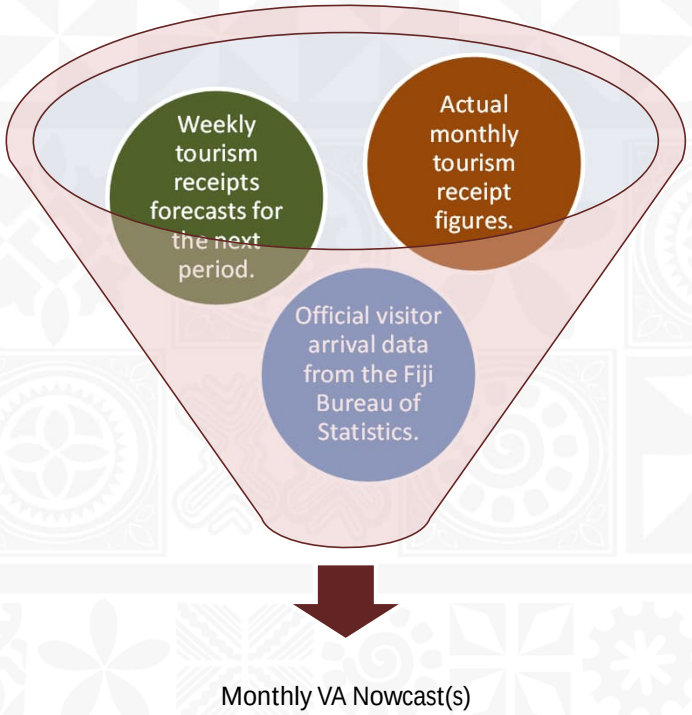
Highly Confidential

Motivation



Highly Confidential

Core data inputs and methodology



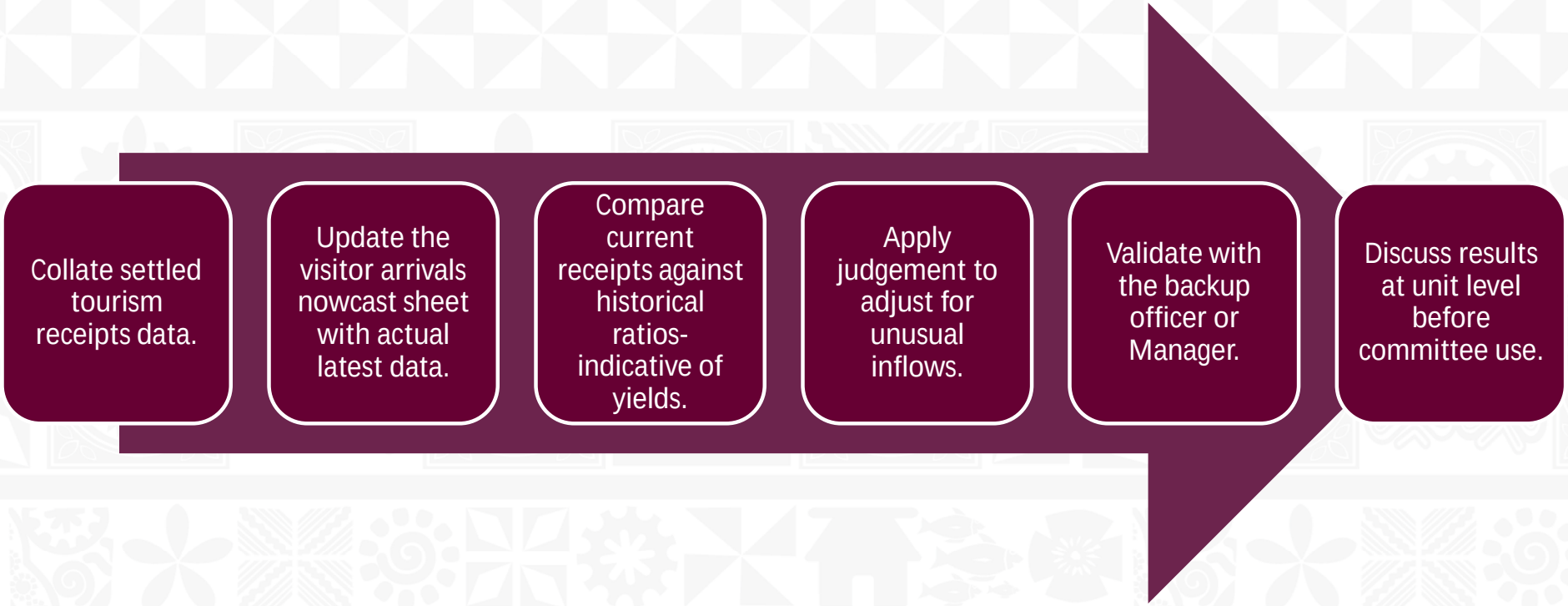
Ratio (January) = Tourism Receipts (January)/ Visitor arrivals (January)

Ratio methods

- T-12: The most recent ratio of the same month
- Two-year (monthly) average
- Minimum ratio
- Maximum ratio
- 12 MA: (12 Month average)

Highly Confidential

The process



Highly Confidential

Role of Judgement

- Overseas Exchange Transaction code 29 includes some non-tourism-related inflows and human error.
- Identify and exclude unusual transactions where relevant.
- Professional judgement improves the reliability of the nowcast.
- The process combines data, ratios, and some contextual interpretation.

Highly Confidential

Out of sample forecasting assessment

Month	Arrivals	t-12	2 yr avg	min coeff	max coeff	12 MA
Jan-24	70.3	68.4	46.9	68.4	35.7	85.0
Feb-24	54.7	53.2	40.5	53.2	32.8	55.8
Mar-24	69.3	60.0	51.8	60.0	45.6	60.1
Apr-24	77.3	94.9	89.1	94.9	84.1	85.3
May-24	82.9	89.6	81.8	89.6	75.3	87.2
Jun-24	92.6	94.7	95.0	95.3	94.7	93.0
Jul-24	98.3	113.7	110.7	113.7	107.9	110.6
Aug-24	94.8	96.4	90.0	96.4	84.3	100.5
Sep-24	92.6	112.3	100.8	112.3	91.4	91.5
Oct-24	89.4	101.2	100.3	101.2	99.5	96.9
Nov-24	76.8	83.8	81.5	83.8	79.2	77.1
Dec-24	83.8	75.8	82.6	90.8	75.8	70.4
Jan-25	70.8	79.5	78.4	79.5	77.3	90.3
Feb-25	49.5	55.6	54.8	55.6	54.0	52.6
Mar-25	63.8	77.1	71.5	77.1	66.7	60.5
Apr-25	80.4	75.6	83.4	92.9	75.6	77.3
May-25	86.6	75.1	78.0	81.1	75.1	74.4
Jun-25	92.7	94.6	95.6	96.7	94.6	89.5
Jul-25	99.3	99.5	106.7	115.1	99.5	106.8
Aug-25	99.7	90.6	91.3	92.1	90.6	92.4
Sep-25	92.3	96.8	106.1	117.4	96.8	93.1
Oct-25	88.3	87.5	92.9	99.0	87.5	93.3
Nov-25	78.0	69.9	72.9	76.3	69.9	70.1
Dec-25	85.0	90.5	85.9	90.5	81.8	75.1
Jan-26	71.0	73.0	77.2	82.0	73.0	92.8
Feb-26	54.2	58.7	62.1	65.9	58.7	62.3
Mar-26	71.8	73.1	80.0	88.2	73.1	70.0
Apr-26	79.7	94.3	91.4	94.3	88.7	89.9
May-26	85.4	92.7	86.1	92.7	80.4	77.6
Jun-26	98.5	110.7	111.8	112.9	110.7	103.4
Jul-26	105.8	112.4	112.5	112.6	112.4	116.3
Aug-26	100.0	113.6	108.1	113.6	103.2	99.5
Total		8.0	5.0	2.0	5.0	12.0

** All values in the table are denominated in thousands*

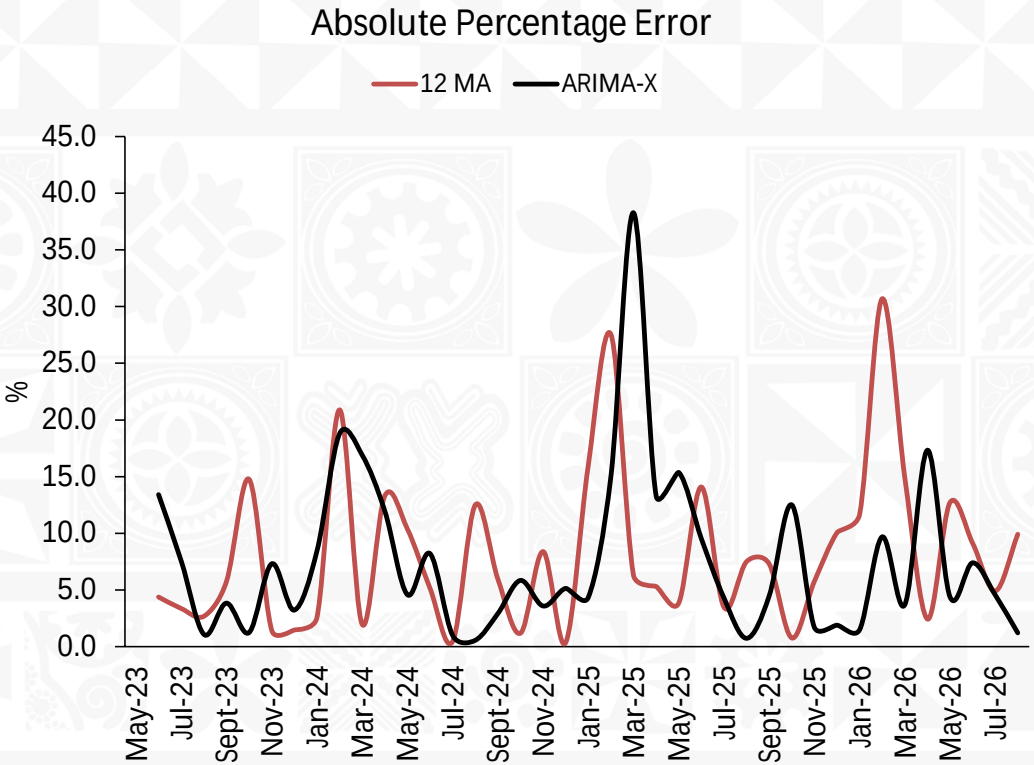
Highly Confidential

32	t-12	2 yr-avg	min coeff	max coeff	12 MA
Avg error	82.7	86.9	128.9	108.3	78.5
RMSE	9.1	9.3	11.4	10.4	8.9

Findings

- 32 months of back cast data indicate that 12 MA and T-12 were the best two estimators
- The variability of the closest estimates support its use as a range forecast
- However, for a point forecast, the 12 MA is the preferred estimate

12 MA nowcast fares well against the ARIMAX but is not better



	RMSE		
Period	Random	12 MA	Nowcast
May 2023-Aug 2026	18.5	8.3	7.0
Jan 2024-Aug 2026	18.5	8.9	7.3

Findings

- Evident upper bias
- First quarter disease
- Excluding Jan-March outcomes, it fares well against the ARIMAX
- Can be improved with a combination forecast

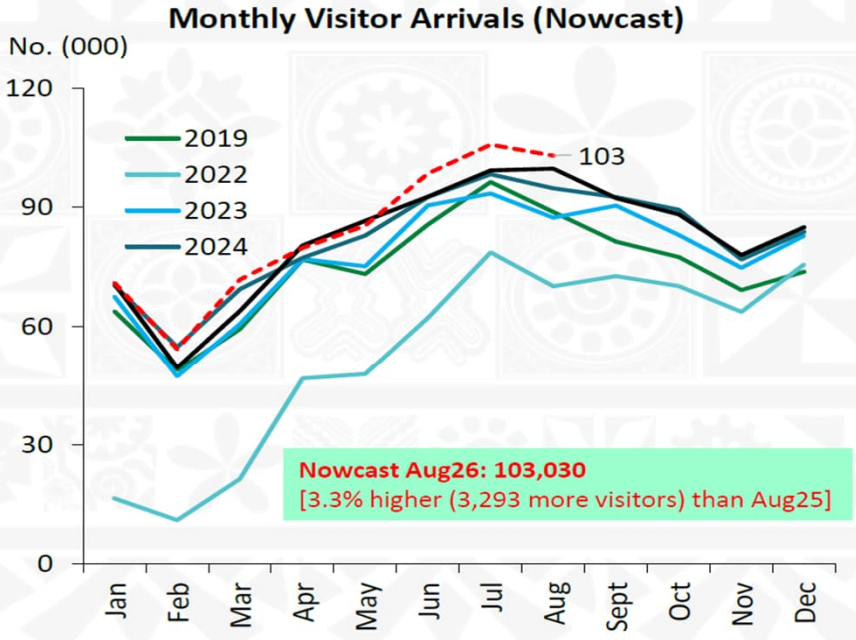
Highly Confidential

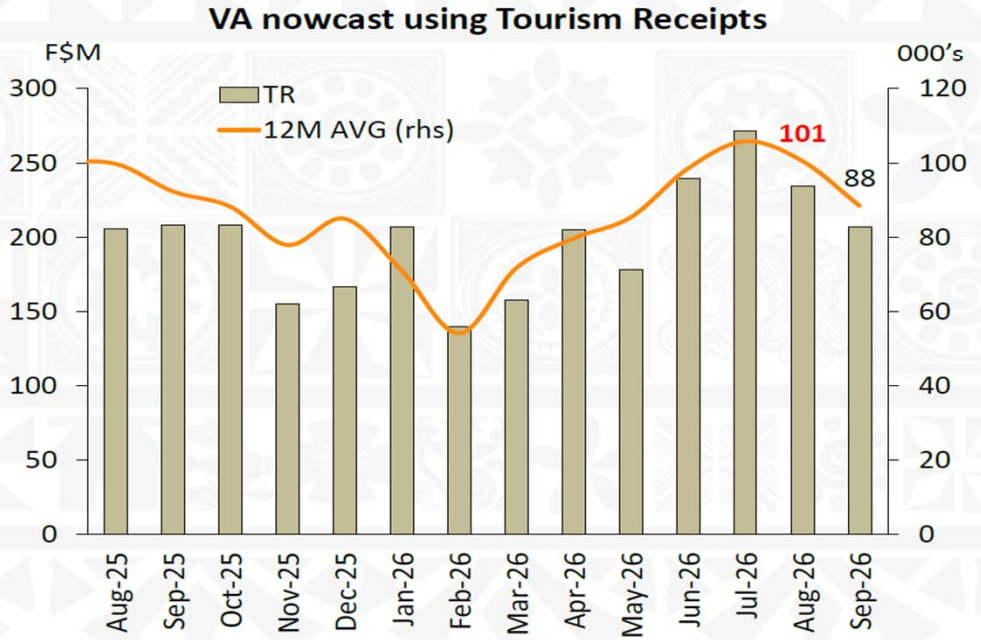
ST1

Communication of the VA Nowcasting Model

Visitor arrivals are nowcasted at 101-103k in August before easing to 88k in September

Risks:
- Upside: 0.5-0.6 ppt higher than 1.7% forecast
- Downside: 5% Tourism Services Tax; cost of living in source countries; competition and global uncertainty





Highly Confidential

Slide 19

ST1 Will paste pics of Board slide (with VA) and BA write-up on VA.
Saimoni Tuiqara, 2026-09-27T08:13:21.862

Linkages to Wider Outputs

1

Monthly foreign reserves analysis.

Weekly foreign reserves forecasting.

2

Monthly VA nowcasting.

3

Cumulative visitor arrivals assessment.

PCC, MPC, and Board presentations.

Monthly economic conditions paper.

Highly Confidential

Risks and Controls

Risk	Data accuracy – While initial estimates use confirmed data, updates use weekly forecasts as assumed values of the month-end figure.	Weekly FR trend assumptions.
Risk Category	Variance in VA nowcasting	Impacts cumulative VA assessment
Risk Rating	Likely	Likely
Risk Treatment	Reduce	Reduce
Control	Understand and discern the real level of foreign card related inflows from the daily trend of tourism receipts. The assumptions must exclude any identified level of other related flows (project financing, etc) that may appear in the narration.	Understand and discern the real level of foreign card related inflows from the daily trend of tourism receipts. The assumptions must exclude any identified level of other related flows (project financing, etc) that may appear in the narration.
Control Type	Preventative	Preventative
Residual Risk	Low	Low

Highly Confidential

Key Takeaways

1. This simple method provides timely insight into visitor arrivals.
2. The approach strengthens real-time assessment of Fiji's tourism sector.
3. The process supports the ARIMAX model for proactive policy advice and timely reporting.
4. The process is linked to wider standard outputs.
5. Validation and judgement are essential to minimise risk.

Highly Confidential

Fiji's GDP Nowcast Model

Highly Confidential

The Current Process

- Two official macroeconomic forecasting rounds annually [Jun & Nov]
- Excel-based forecasting
  - ✓ GDP Production Method
  - ✓ Forecast more than 182 lines of economic sector activities
  - ✓ Currently 2019 GDP base
- Joint forecasts by the Reserve Bank of Fiji and Ministry of Finance, reviewed by technical and executive committees
- Lagged GDP data: GDP actual last year released by FBOS at H2 of current year

Highly Confidential

Fiji's GDP Current Forecasting Process- Policy Use

Fiji's National Budget document



Assess debt as a percent of GDP



Policy Deliberation *RBF Board, MPC, PCC*



External Liaison/Presentation

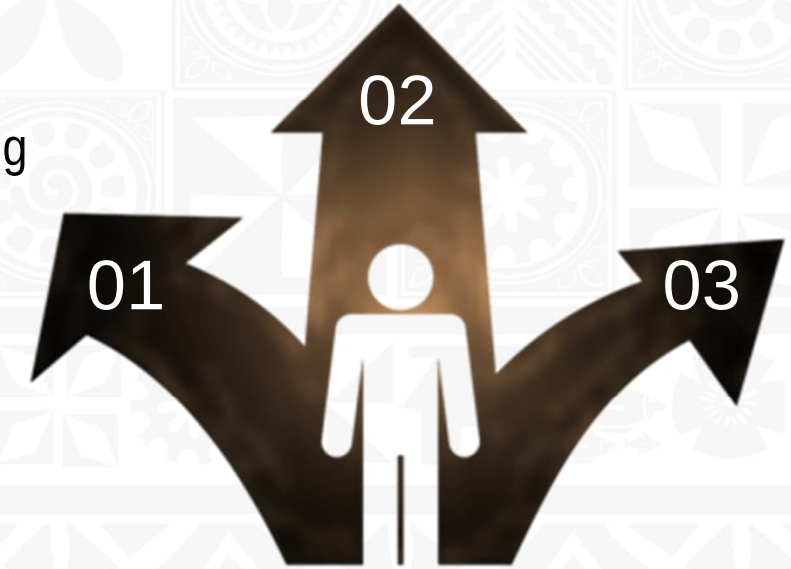


Highly Confidential

Fiji's GDP Current Forecasting Process-Opportunities

Reduce time-intensive forecasting through automated nowcasting models

Explore GDP forecasting using expenditure or income approach



Reduce forecast bias by complementing expert judgment with model-based analysis

Background/Context

- Developed following a training conducted by Mr Hayk (PFTAC Advisor)- with RBF's GDP officer and other ECG staff
- Model was developed around February 2026 with variables selected and discussed during the training
- Model is estimated on Eviews and can also be replicated using Python

- <u>Current use-</u> Model is in testing and refinement phase- results are currently monitored by RBF's DC team.



Highly Confidential

The Nowcasting Model

Highly Confidential

Model Specification

Fiji's GDP

- Annual Frequency

Independent Variables

- Have varying frequencies and data collection years

Data Transformation

- Summation Seasonal Adjustment, Interpolation conducted using Eviews

Sample

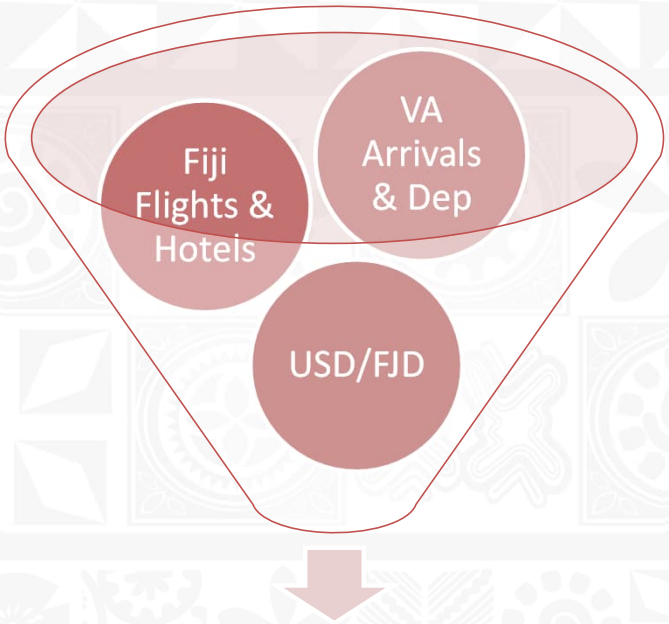
- 1992 -2024

Variables Used

- Flow : Indirect tax (VAT collection), Hotel turnover
- Stock : VAT rate, GDP

Highly Confidential

Variable Selection



Initial Pool of Variables

Variable Selection

- ✓Correlation test analysis b/w Fiji's GDP and each potential determination
- ✓Prior Economist knowledge and experience needed to make sense of variables



Final Variables

- ✓Real indirect taxes
- ✓Hotel turnover
- ✓VAT rate

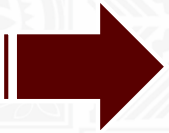
Highly Confidential

Nowcasting in Action

Variables were fitted together using Bridge estimation method



Forecast independent variables using the EViews Auto Arima function



Combined forecasts with actual data and annualized the series



Annual series are then used to nowcast the current GDP forecast for Fiji for 2026.

Highly Confidential

Nowcasting in Action- Model Equation and Output

$$RGDP_t = constant + DLOG\left(RGDP\ (-1)\right) + DLOG\ (Real\ Indirect\ Tax) + DLOG\ (Real\ Hotel\ Earnings) + D\ (VAT\ Rate)$$

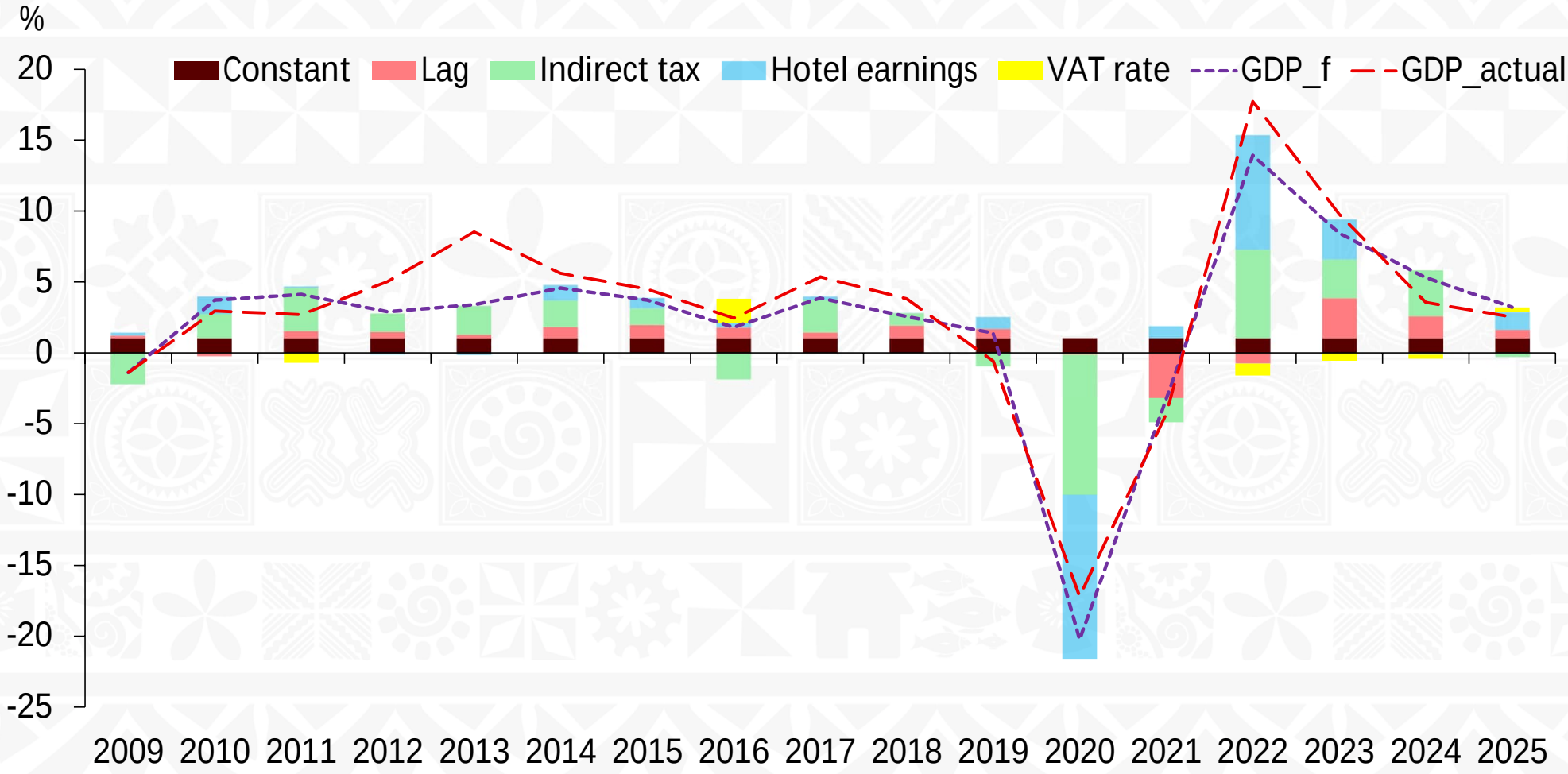
Dependent Variable: DLOG(R_GDP)
Method: Least Squares
Date: 09/18/26 Time: 10:15
Sample (adjusted): 2006 2024
Included observations: 19 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.010202	0.003734	2.732016	0.0162
DLOG(R_GDP(-1))	0.180019	0.062272	2.890853	0.0119
DLOG(R_N_TAX_INDIRECT)	0.162728	0.044098	3.690135	0.0024
DLOG(R_HOTEL_EARN)	0.082978	0.017923	4.629682	0.0004
D(VAT_RATE_Q)	-0.159803	0.228509	-0.699329	0.4958

R-squared	0.957334	Mean dependent var	0.020208
Adjusted R-squared	0.945144	S.D. dependent var	0.066686
S.E. of regression	0.015619	Akaike info criterion	-5.259759
Sum squared resid	0.003415	Schwarz criterion	-5.011222
Log likelihood	54.96771	Hannan-Quinn criter.	-5.217696
F-statistic	78.53345	Durbin-Watson stat	2.283666
Prob(F-statistic)	0.000000		

Highly Confidential

Nowcasting GDP Results (Decomposition)



Highly Confidential

Nowcasting Model- Current Phase

- Internal unit/Group training and knowledge sharing
- Forecast validation/ variable scrutiny phase
- Strategising way forward (Communication/ Policy Use)

Highly Confidential

Nowcasting GDP Model- Challenges

- Staffing and Resources;
- Technical Capacity;
- Less comprehensive- results can be subject to more scrutiny;
- Models can be less intuitive from the excel based, more statistical;
- Causal vs correlation interpretation;
- Data limitation Gaps

Highly Confidential

Nowcasting Model- Potential Policy Use (Way Forward)

- Monthly Policy and Board meetings- update
- Bi-annual GDP stocktake comparison analysis
- Alternative forecast to the Macroeconomic Committee forecast

Highly Confidential

Nowcasting Model- Communication

- Similar to VA Nowcast reporting- Results can be communicated through monthly Board publications (1–2-page summary);
- Presentation through Policy and Board slides;
- Presentation through Macroeconomic Technical Committee meeting and potential use in official press release and National Budget write-up.

Highly Confidential


VINAKA